14 Lecture 14

14.1 Types of convergence of random variables

For a sequence $(X_n)_{n\in\mathbb{N}}$ of random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with values in $\mathbb{R}$, we know from calculus two basic modes of convergence: pointwise convergence and uniform convergence. In this chapter we will define some new, natural notions of convergence with respect to which a sequence might converge more easily, and which are moreover useful in applications of Probability to Statistics and elsewhere.

The notions of convergence that we will discuss are: almost sure convergence, L^p convergence and convergence in probability. We first define the space L^p.

Definition 14.1. Let $p \geq 1$. We say that a random variable $X \in L^p$ if $\mathbb{E}[|X|^p] < \infty$.
L^p is a vector space, that is, if $X, Y \in L^p$, then $X + aY \in L^p$ for any $a \in \mathbb{R}$.

Note that we have already seen L^p spaces when $p = 1$ (integrable random variables) and $p = 2$ (square integrable random variables or random variables with finite variance). We are now ready to define the various types of convergence.

Definition 14.2. Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables as above.

1. We say that $(X_n)_{n\in\mathbb{N}}$ converges to a random variable X *with probability one* or *almost surely*, and we write $X_n \xrightarrow{\text{a.s.}} X$, if
$$\mathbb{P}\Big(\lim_{n\to\infty} X_n = X\Big) = 1, \qquad \text{that is,} \qquad \mathbb{P}\Big(\Big\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\Big\}\Big) = 1.$$
2. For $p \geq 1$ and $X_n, X \in L^p$ for every $n \in \mathbb{N}$, we say that $(X_n)_{n\in\mathbb{N}}$ converges to X in L^p, and we write $X_n \xrightarrow{L^p} X$, if
$$\lim_{n\to\infty} \mathbb{E}[|X_n - X|^p] = 0.$$
3. We say that $(X_n)_{n\in\mathbb{N}}$ converges to X *in probability*, and we write $X_n \xrightarrow{\mathbb{P}} X$, if
$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0, \quad \text{for every } \varepsilon > 0.$$

Example 1
Consider the probability space $([0,1], \mathcal{B}([0,1]), \lambda)$, where λ is the Lebesgue measure. Let's study the convergence of the sequence random variables $(X_n)_{n\in\mathbb{N}}$ with
$$X_n(\omega) = \begin{cases} n, & \text{if } \omega \in (0, 1/n), \\ 0, & \text{if } \omega \in [0,1] \setminus (0, 1/n). \end{cases}$$

For any fixed $\omega \in [0,1]$, it is easy to see that $X_n(\omega) \xrightarrow{n\to\infty} 0$, hence $X_n \xrightarrow{\text{a.s}} 0$. We also have $X_n \xrightarrow{\mathbb{P}} 0$, since for any $\varepsilon > 0$
$$\mathbb{P}(|X_n| > \varepsilon) \leq \mathbb{P}(\{\omega \in \Omega : \omega \in (0, 1/n)) = \lambda((0, 1/n)) = \frac{1}{n} \xrightarrow{n\to\infty} 0.$$

However, for any $p \geq 1$, $X_n \not\xrightarrow{L^p} 0$, because
$$\mathbb{E}[|X_n|^p] = n^p \mathbb{P}(X_n = n) + 0^p \mathbb{P}(X_n = 0) = n^p \cdot \frac{1}{n} = n^{p-1} \not\to 0.$$

If we know that one of the convergences holds, we might be able to get some information about the others. We prove the following theorems.

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Theorem 14.3. *Suppose that $X, (X_n)_{n\in\mathbb{N}}$ are random variables and $p \geq 1$.*

1. *If $X_n \xrightarrow{L^p} X$, then $X_n \xrightarrow{\mathbb{P}} X$.*
2. *If $X_n \xrightarrow{a.s} X$, then $X_n \xrightarrow{\mathbb{P}} X$.*

Proof. Let $\varepsilon > 0$.

1. By Markov's inequality, we have
$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(|X_n - X|^p > \varepsilon^p) \leq \frac{1}{\varepsilon^p}\mathbb{E}[|X_n - X|^p] \xrightarrow{n\to\infty} 0.$$
2. We observe that
$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{E}[\mathbf{1}_{|X_n - X|>\varepsilon}].$$
Since $X_n \xrightarrow{\text{a.s}} X$, we have $\mathbf{1}_{|X_n-X|>\varepsilon} \xrightarrow{\text{a.s, } n\to\infty} 0$. In addition, $\mathbf{1}_{|X_n-X|>\varepsilon}$ is a bounded sequence of random variables, therefore the dominated convergence theorem yields
$$\lim_{n\to\infty} \mathbb{E}[\mathbf{1}_{|X_n-X|>\varepsilon}] = \mathbb{E}[\lim_{n\to\infty} \mathbf{1}_{|X_n-X|>\varepsilon}] = 0.$$
The proof is complete.

□

Theorem 14.4. *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables such that $X_n \xrightarrow{\mathbb{P}} X$. Then there exists a subsequence $(X_{n_k})_{k\in\mathbb{N}}$ such that $X_{n_k} \xrightarrow{a.s,\ k\to\infty} X$.*

Proof. Since $X_n \xrightarrow{\mathbb{P}} X$, for any $k \in \mathbb{N}$, we may choose $n_k \in \mathbb{N}$ such that
$$\mathbb{P}\left(|X_{n_k} - X| > \frac{1}{k}\right) < \frac{1}{2^k}.$$
We set $A_k = \{\omega \in \Omega : |X_{n_k}(\omega) - X(\omega)| > 1/k\}$. We notice, by Tonelli's theorem, that
$$\mathbb{E}\left[\sum_{k=1}^{+\infty} \mathbf{1}_{A_k}\right] = \sum_{k=1}^{+\infty} \mathbb{E}[\mathbf{1}_{A_k}] = \sum_{k=1}^{+\infty} \mathbb{P}(A_k) \leq \sum_{k=1}^{+\infty} \frac{1}{2^k} = 1 < \infty.$$
By Example 1 from Lecture 12, this implies $\sum_{k=1}^{+\infty} \mathbf{1}_{A_k} < \infty$ almost surely, therefore with probability 1 at most finitely many of the events A_k happen. This means that $|X_{n_k}(\omega) - X(\omega)| \leq 1/k$ for k large enough, which means $|X_{n_k}(\omega) - X(\omega)| \xrightarrow{k\to\infty} 0$ or $X_{n_k} \xrightarrow{\text{a.s}} X$. □

Theorem 14.5. *Let $p \geq 1$ and $X, (X_n)_{n\in\mathbb{N}}$ as in Theorem 14.4 with the extra assumption that $|X_n| \leq Y$ for some $Y \in L^p$. Then, $X \in L^p$ and $X_n \xrightarrow{L^p} X$.*

Proof. We know from Theorem 14.4 that $X_{n_k} \xrightarrow{\text{a.s}} X$ for some subsequence $(X_{n_k})_{k\in\mathbb{N}}$. We have by Fatou's lemma
$$\mathbb{E}[|X|^p] = \mathbb{E}[\lim_{k\to+\infty} |X_{n_k}|^p] \leq \liminf_k \mathbb{E}[|X_{n_k}|^p] \leq \mathbb{E}[|Y|^p] < \infty,$$
hence $X \in L^p$. For the second implication we argue by contradiction. Assume that there exists $\varepsilon > 0$ and subsequence $(X_{\lambda_k})_{k\in\mathbb{N}}$ that
$$\mathbb{E}[|X_{\lambda_k} - X|^p] \geq \varepsilon.$$
By the convergence in probability, we know that there is a further subsequence $(X_{\lambda_{n_k}})_{k\in\mathbb{N}}$ such that $X_{\lambda_{n_k}} \xrightarrow{\text{a.s}} X$. We have
$$|X_{\lambda_{n_k}} - X|^p \leq (|X_{\lambda_{n_k}}| + |X|)^p \leq (2|Y|)^p = 2^p|Y|^p \quad \text{and} \quad \mathbb{E}[2^pY^p] < \infty,$$
so by the Dominated Convergence Theorem,
$$\varepsilon < \lim_{k\to\infty} \mathbb{E}[|X_{\lambda_{n_k}} - X|^p] = 0,$$
which is a contradiction. □

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We see that convergence in L^p (for $p \geq 1$) requires an additional assumption. Note that, by Example 1, almost sure convergence does not in general imply L^p convergence. However, since almost sure convergence implies convergence in probability, the previous theorem states that if

$$X_n \xrightarrow{\text{a.s.}} X \quad \text{and there exists } Y \in L^p \text{ such that } |X_n| \leq Y \text{ for all } n \in \mathbb{N},$$

then $X \in L^p$ and $X_n \xrightarrow{L^p} X$.

Example 2 (L^1 vs L^2 convergence)
Assume that $X_n \xrightarrow{L^2} X$. Let's show that $X_n \xrightarrow{L^1} X$.

Indeed, by Cauchy-Schwarz we have

$$\mathbb{E}[|X_n - X|] \leq \sqrt{\mathbb{E}[|X_n - X|^2]} \xrightarrow{n\to\infty} 0.$$

The converse is not true. For example, consider the sequence $(X_n)_{n\in\mathbb{N}}$ in the probability space $(\Omega, \mathcal{F}, \mathbb{P}) = ((0,1], \mathcal{B}((0,1]), \lambda)$ with

$$X_n(\omega) = \sqrt{n}\,\mathbf{1}_{(0,\,1/n]}(\omega), \qquad \omega \in (0,1].$$

We have

$$\mathbb{E}[|X_n|] = \sqrt{n}\lambda((0,1/n]) = \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} \xrightarrow{n\to\infty} 0 \quad \text{and} \quad \mathbb{E}[|X_n|^2] = 1,$$

hence $X_n \xrightarrow{L^1} 0$ but $X_n \not\xrightarrow{L^2} 0$.

Continuous functions preserve almost sure convergence and convergence in probability.

Theorem 14.6. *Let $f\colon \mathbb{R} \to \mathbb{R}$ be a continuous function and $(X_n)_{n\in\mathbb{N}}, X$ random variables.*

1. *If $X_n \xrightarrow{a.s.} X$, then $f(X_n) \xrightarrow{a.s.} f(X)$.*
2. *If $X_n \xrightarrow{\mathbb{P}} X$, then $f(X_n) \xrightarrow{\mathbb{P}} f(X)$.*

Proof. 1. We consider the sets

$$A = \{\omega \in \Omega : X_n(\omega) \xrightarrow{n\to\infty} X(\omega)\},$$
$$B = \{\omega \in \Omega : f(X_n(\omega)) \xrightarrow{n\to\infty} f(X(\omega))\}.$$

For almost any $\omega \in A$, we have $X_n(\omega) \xrightarrow{n\to\infty} X(\omega)$, thus $f(X_n(\omega)) \xrightarrow{n\to\infty} f(X(\omega))$, that is $\omega \in B$. This means that $A \subset B \implies 1 = \mathbb{P}(A) \leq \mathbb{P}(B) \implies \mathbb{P}(B) = 1$.

2. Let $\varepsilon > 0$ and $M > 1$. Since f is continuous, it must be uniformly continuous in $[-M, M]$, hence there exists a $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $x, y \in [-M, M]$ and $|x - y| < \delta$. We have

$$\begin{aligned}\mathbb{P}(|f(X_n) - f(X)| > \varepsilon) &\leq \mathbb{P}(\{|X_n - X| < \delta, |X| > M - \delta\} \cup \{|X_n - X| > \delta\}) \\ &\leq \mathbb{P}(|X| > M - \delta) + \mathbb{P}(|X_n - X| > \delta).\end{aligned}$$

Letting $n \to +\infty$ we derive

$$\limsup_n \mathbb{P}(|f(X_n) - f(X)| > \varepsilon) \leq \mathbb{P}(|X| > M - \delta).$$

Finally, letting $M \to +\infty$ yields

$$\limsup_n \mathbb{P}(|f(X_n) - f(X)| > \varepsilon) \leq \mathbb{P}(|X| > +\infty) = 0,$$

hence $\lim_n \mathbb{P}(|f(X_n) - f(X)| > \varepsilon) = 0$.

□

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